

- 6 (a) Electromotive force of a source is defined as the **energy converted per unit charge from other forms of energy into electrical energy** when the charge is driven through the source round a complete circuit. 1

Potential difference between two points of a circuit **is the amount of electrical energy converted per unit charge into other forms of energy** when the charge moved from one point to the other. 1

- (b) (i) Using $P = I^2 R$ 1

$$320 \, \Omega : \text{current} = \sqrt{\frac{0.8}{320}} = 0.050 \, \text{A} \quad 1$$

$$500 \, \Omega : \text{current} = \sqrt{\frac{P}{R}} = \sqrt{\frac{0.8}{500}} = 0.040 \, \text{A} \quad 1$$

- (ii) Max current in parallel branch = $2 \times 0.040 = 0.080 \, \text{A}$

But the $320 \, \Omega$ resistor will be damaged first.

Hence, maximum current in circuit = $0.050 \, \text{A}$ 1

1 mark for recognizing the maximum current or the resistor which will be damaged first.

Maximum safe p.d = $0.050 (320 + 250) = 28.5 \, \text{V}$

1 mark for total resistance 1

1 mark for determining the maximum safe potential difference. 1

- 7 (a) **The root-mean-square (r.m.s.)** voltage of an alternating current is